

Summary

AI Engineer with 2 years of experience across AI and software engineering, focused on designing and building production-grade agentic systems, RAG pipelines, LLM integrations, and NLP solutions. Master's in Computer Science with specialisation in AI. Holder of a Dutch Orientation Year (Zoekjaar) visa, available from June 2026.

Work Experience

AI Engineer

Pet 24-7 Global Ltd

July 2025 – Present

London, United Kingdom

 Live app: App Store – Google Play

Agentic RAG Platform

- Built the chatbot as a **LangGraph ReAct agent** with pluggable tools for knowledge base retrieval, semantic history search, multimodal OCR, image generation, web search, and PDF generation — enabling autonomous multi-step veterinary consultations.
- Engineered a **hybrid memory architecture** combining a PostgreSQL session cache with Milvus vector search for both fast recent-turn recall and long-term semantic history retrieval.
- Integrated **Langfuse** for end-to-end LLM observability across all agent steps and tool calls.
- Built **MCP servers** to expose agent tools as modular, standardized interfaces.

Production RAG System

- Built and deployed a **production RAG veterinary chatbot** integrating semantic retrieval, cross-encoder re-ranking, and multilingual query handling over a domain-specific knowledge base.
- Designed a **per-user episodic memory system** using vector embeddings to store and semantically retrieve past clinical cases, enabling context-aware follow-up responses across sessions.
- Delivered **real-time SSE streaming** and a multimodal pipeline covering OCR, speech-to-text, and PDF generation.

RAG Retrieval Benchmark

- Benchmarked **Milvus, Qdrant, and Weaviate** across chunking strategies (recursive, semantic, fixed-size), evaluating latency, throughput, and retrieval quality.

AI Analytics Dashboard

- Built an end-to-end **NLP analytics pipeline** processing all chatbot conversations through a two-layer NER system (domain dictionary + biomedical token classifier) and BERTopic-based topic modeling with LLM-generated labels.
- Developed an **AI-powered analytics dashboard** providing real-time NLP insights — entity trends, topic modeling, co-occurrence analysis, and regional breakdowns — via a React frontend and FastAPI backend.

Full Stack Developer

MeNEW - Personal Digital Menu

Nov 2023 – Nov 2024

Part-time

Digital Menu Platform

- Built and launched a full-stack digital menu platform using Next.js (TypeScript), Django REST API, and GraphQL, with CI/CD pipelines for automated testing and deployment.

Technical Skills

Languages	Python, Java, C++ , JavaScript, TypeScript
AI/ML	NLP, LLM, RAG, Agentic AI, Deep Learning, Computer Vision, Classification, Regression, Clustering
Frameworks	PyTorch, TensorFlow, Hugging Face, LangChain, LangGraph, FastAPI, Keras, Spark
Libraries	Pandas, NumPy, NLTK, SciPy, BERTopic, Matplotlib, Plotly, Seaborn
Vector Databases	Milvus, Qdrant, Weaviate, ChromaDB
Cloud & Storage	AWS S3, Google Cloud (Vertex AI)
LLM APIs	Google Gemini, OpenAI API, MCP
Web & DevOps	React, Next.js, Node.js, Django, Docker, Git, SQL, MongoDB, PostgreSQL, Prisma, JWT

Education

University of Padua
Master in Computer Science, Specialisation in Artificial Intelligence
GPA: **105/110**

Padova, Italy
Sep 2022 – Sep 2025

Thesis: Improving Navigation in LLM-Based AI Frameworks

October 2024 – July 2025

Designed and built an agentic AI framework where an LLM acts as the reasoning engine, dynamically selecting and composing domain-specific tools at runtime via structured APIs and in-context prompting. The system follows a **ReAct-style reasoning loop** — generating executable code, observing intermediate results, and iterating toward a solution. Benchmarked against end-to-end and modular baselines to evaluate performance and robustness.

Supervisor: Prof. Lamberto Ballan

Course Projects

- **Sentiment Analysis and Topic Modeling with Fine-tuned Transformer Models** (*Code and report on GitHub*)
- **Image Colorization with Conditional GANs and Pretrained U-Net** (*Code and report on GitHub*)
- **CNN and Vision Transformer Fine-tuning for Object Detection** (*Code and report on GitHub*)
- **LLM Pruning and Fine-Tuning for Efficient Text Summarization**

For other works, explore more on my *GitHub*

Languages

English: C2

Dutch: A2

Italian: A2

Persian: Native